

Programming for AI (AI-2001)

Sessional-I Exam

Total Time (hrs): 01

Total Marks: 15

Total Questions: 03

Date:

Time:

Course Instructor

Mr. Talha Shahid

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Student Name	Roll No	Section	Student Signature

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CLO # 1: Comprehended the fundamental construct of the python for data analysis and representation using data structures like **Lists** and **Dictionaries**.

Q1:

[wgt: 04]

You've been hired as a junior developer for "**Karachi Readers**," a popular local bookstore. They need a simple program to help manage their inventory. The manager has given you a small list of their current best-sellers to start with.

The current inventory is as follows:

Book Title	Author	Price (PKR)	Stock
The Alchemist	Paulo Coelho	1200	15
Forty Rules of Love	Elif Shafak	1500	25
Sapiens	Yuval Noah Harari	2200	10
Atomic Habits	James Clear	1800	30

Write a Python script to help the manager with the following tasks:

1. **Represent the Inventory:** First, store the bookstore's inventory data within your program.
2. **Check Stock:** Create a feature that allows the manager to input a **Book Title** and see the current stock **count** for that book. If the book is not in the inventory, it should display a message saying, "**Book not found.**"
3. **Calculate Total Value:** Calculate the total **monetary value** of the entire inventory. This is the sum of the price of each book multiplied by its stock quantity.

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4. **Generate a Reorder List:** The manager needs a list of all books that have a stock count of 15 or less. Display the titles of these books so they know what to reorder.

Possible Output(s)	
<pre>--- Karachi Readers' Inventory Tool --- CHECK BOOK STOCK Enter the title of the book to check: Atomic Habits Stock for 'Atomic Habits': 30 copies available. CALCULATE TOTAL INVENTORY VALUE The total value of all books is: PKR 131,500.00 GENERATE REORDER LIST (15 or fewer copies) The Alchemist (Stock: 15) Sapiens (Stock: 10)</pre>	<pre>--- Karachi Readers' Inventory Tool --- CHECK BOOK STOCK Enter the title of the book to check: my book 'Book not found.' CALCULATE TOTAL INVENTORY VALUE The total value of all books is: PKR 131,500.00 GENERATE REORDER LIST (15 or fewer copies) The Alchemist (Stock: 15) Sapiens (Stock: 10)</pre>

CLO # 1: Comprehended the fundamental construct of the python for data analysis and representation using data structures like **Lists**, **Dictionaries** and modular programming using custom and in-built **Functions**.

Q2:

[wgt: 04]

You're a developer for a logistics company, "**Karachi Quick Ship**." A delivery truck driver needs a simple tool to find the **nearest warehouse** to their **current location** to pick up a new shipment. The locations are all mapped on a **simple 2D grid**.

Your task is to build a command-line tool that **repeatedly asks** the driver for their **location** and tells them which **warehouse is closest**.

Given Information:

- **Warehouse Locations:** A list of (x, y) coordinates for all available warehouses.

$$warehouses = (2, 8), (9, 1), (12, 11), (1, 2), (7, 6)$$

- **Distance Formula:** The distance d between two points (x_1, y_1) and (x_2, y_2) is calculated using the Euclidean distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Possible Output
<pre>--- Karachi Quick Ship's Warehouse Finder --- Enter your current X and Y coordinates or type 'quit' to exit. Note: This version assumes you will enter valid numbers for coordinates. Enter your X coordinate: 7 Enter your Y coordinate: 9 The nearest warehouse is at [7, 6] with a distance of 3.00 units. Enter your X coordinate: 25 Enter your Y coordinate: 32 The nearest warehouse is at [12, 11] with a distance of 24.70 units.</pre>

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Enter your X coordinate: 15
Enter your Y coordinate: 9
The nearest warehouse is at [12, 11] with a distance of 3.61 units.

Enter your X coordinate: quit
Exiting tool. Safe travels!

CLO # 2: Understand and apply the **Object-Oriented** concepts in the python.

Q3:

[wgt: 07]

You are tasked with developing a simple system to model the structure of a university. The system needs to represent **departments**, the **courses** they offer, and the **professors/lecturers** who teach them.

Here's how the system should work logically:

1. A **Course**, like "**Programming for AI**," is created by and belongs **exclusively** to a **specific Department**. A course **shouldn't exist** in the system **without** a department that offers it.
2. A **Faculty**(professor/lecturer) is an employee who can be assigned to teach in a department. They are **independent individuals** who can be created in the system first and then assigned to a department **later**.

Write a Python script using **Object-Oriented Programming** to model this scenario by creating the following classes and methods.

Class	Member	Description
Faculty	name	Stores the professor's/lecturer's full name.
	employeeID	Stores the professor's/lecturer's unique ID.
Course	courseCode	Stores the unique code for the course (e.g., "AI-2001").
	courseTitle	Stores the full title of the course.
Department	name	Stores the name of the department.
	courses	A list to hold the Course objects by the department.
	faculty	A list to hold references to assigned Professor/Lecturers objects.
	addCourse(para:)	Creates a new Course object and adds it to the courses list with course code and course title.
	assignProfessor para:)	Adds a pre-existing Professor/Lecturer to the faculty list.
	displayDetails(para:)	Prints all formatted details for the department.

Possible Output

```
Department: Computer Science
--- Courses Offered ---
- DS-5007: Natural Language Processing
- AI-2001: Programming for AI
--- Assigned Faculty ---
- Dr. Muhammad Rafi (ID: EMP000)
- Mr. Talha Shahid (ID: EMP001)
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